

Trace — One Honest Sustainability Number Per Batch

A brutally simple sustainability tracker built for Indian SME manufacturers.

Trace Banner

 **Live Demo**

 **Try the App**

<https://trace-a-brutally-simple-sustainabil.vercel.app/>

 **GitHub Repository**

<https://github.com/Trinu15/Trace-A-Brutally-Simple-Sustainability-Tracker-for-Indian-SMEs>

Problem Statement

Small and medium manufacturers in India still manage operational sustainability data across:

- handwritten registers
- electricity bills
- WhatsApp invoices
- paper logs
- storekeeper notebooks

When buyers ask:

“How much water was used per unit for this production batch?”

most factories cannot answer quickly, confidently, or with proof.

Trace solves this problem by converting scattered factory-floor data into:

 **One honest, auditable sustainability score per production batch.**

✨ What Makes Trace Different

Most sustainability tools are:

- enterprise-focused
- dashboard-heavy
- difficult to use
- designed for sustainability officers

Trace is designed for:

- storekeepers
- accounts assistants
- production supervisors
- factory operators

The product focuses on:

- ✓ simplicity
 - ✓ trust
 - ✓ speed
 - ✓ mobile usability
 - ✓ transparent calculations
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Core Features

1 Batch Setup

Create a new production batch in under 2 minutes.

Fields:

- Batch ID
- Product Type
- Number of Units
- Start Date
- Primary Raw Material

Built specifically for low-friction factory-floor usage.

2 Resource Logging

Track operational resource consumption against each batch.

Supported Resource Types

Water

- litres consumed
- source tracking
- municipal
- borewell
- tanker

Electricity

- kWh consumed
- solar contribution tracking

Fuel / Thermal

- diesel
- LPG
- coal

Waste

- solid waste quantity
- disposal method
- landfill
- recycler
- composted
- unknown

Supports:

- multiple entries per batch
- dated entries
- optional operational notes

3 Batch Report Card

When a batch is completed, Trace generates a printable report card showing:

Included Metrics

-  Total water consumption
-  Water per unit
-  Electricity consumption
-  Electricity per unit
-  Fuel consumption

- ✓ Fuel per unit
- ✓ Waste generated
- ✓ Waste per unit

Operational Insights

Each report includes:

- comparison with previous batch
- deduction breakdown
- plain-language operational observation

Example:

“Water consumption per unit increased 18% compared to your previous garment batch.”

4 Honest Sustainability Score

At the top of every report card:

Sustainability Score / 100

The scoring formula is implemented EXACTLY as specified in the challenge.

Scoring Formula

Start at **100**.

Deductions

Condition	Deduction
Water per unit above 50L	1 point per 10L
Electricity per unit above 2kWh	1 point per 0.5kWh
Any fuel usage	-10
Landfill / unknown waste	-15
Recycler waste	-5
Tanker water used	-10

Final Score

- Minimum: 0
- Maximum: 100

No gamification. No badges. No AI-generated scoring.

Just transparent operational math.



A4-Ready PDF Export

Trace generates:

- ✓ clean printable PDF reports
- ✓ A4 optimized layout
- ✓ readable black-and-white formatting
- ✓ procurement-friendly structure

The report card is designed to feel like:

a factory audit document — not a dashboard screenshot.



Trend Dashboard

The dashboard provides:

- sustainability score trend line
- batch history
- top-performing batches
- lowest-performing batches

No projections. No forecasting. Only operational history.



Mobile-First Experience

Trace is optimized specifically for:

- low-end Android devices
- Chrome browser
- 4G internet conditions
- factory-floor usability

UX Priorities

- ✓ Large touch targets
 - ✓ Minimal typing
 - ✓ Fast loading
 - ✓ Simple navigation
 - ✓ Low cognitive load
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Tech Stack

Category	Technology
Frontend	Lovable + React
Styling	Tailwind CSS
Charts	Recharts
Hosting	Vercel
Version Control	GitHub

Design Philosophy

Trace intentionally avoids:

- ✗ ESG complexity
- ✗ enterprise-heavy workflows
- ✗ unnecessary analytics
- ✗ greenwashing visuals
- ✗ gamification

Instead, it focuses on:

“Operational sustainability tracking for real factories.”

Validation

Trace was tested with a real manufacturing / operations user.

Key Validation Insights

- Users understood “batch” terminology immediately
 - Numeric forms were easier than dropdown-heavy flows
 - Transparent deductions increased trust in the score
 - Printable reports felt useful for buyer communication
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Demo Flow

1. Create Batch
 2. Log Resources
 3. Complete Batch
 4. Generate Sustainability Score
 5. Export A4 PDF
 6. View Trend Dashboard
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Performance Goals

- ✓ Loads under 3 seconds
 - ✓ Mobile-first UI
 - ✓ No login required for demo
 - ✓ Preloaded demo factory
 - ✓ Working score engine
 - ✓ A4 PDF support
 - ✓ Real operational workflow
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Suggested Screenshots

Add screenshots for:

- Landing Page
 - Batch Setup
 - Resource Logging
 - Sustainability Report Card
 - PDF Export
 - Trend Dashboard
 - Mobile View
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Project Structure

```
Trace/  
├─ src/  
├─ components/  
├─ pages/  
├─ assets/  
├─ public/  
├─ README.md  
└─ package.json
```

Why This Project Matters

Indian SMEs are increasingly pressured by buyers to provide sustainability metrics.

Most small factories:

- lack dedicated sustainability teams
- rely on paper-based records
- cannot afford enterprise ESG tools

Trace bridges this gap with:

simple, fast, auditable sustainability tracking.

Future Scope

Potential future improvements:

- multilingual support
- OCR bill scanning
- offline syncing
- supplier-level reporting
- utility integrations

(Not included in this submission to maintain strict scope discipline.)



Hackathon Submission Checklist

- ✓ Live URL
 - ✓ Public GitHub Repository
 - ✓ Demo Factory Data
 - ✓ Working Sustainability Formula
 - ✓ Mobile Responsive UI
 - ✓ PDF Export
 - ✓ Validation Recording
 - ✓ Demo Video
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Built For

Indian SME manufacturers who need:

- sustainability visibility
 - buyer-ready reporting
 - operational simplicity
 - trustworthy numbers
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License

MIT License



Final Thought

Trace does not try to be a massive ESG platform.

It solves one real problem exceptionally well:

Giving manufacturers one honest sustainability number they can defend with confidence.